

Community One — DEMO envelope report

Slice: DEMO-SLICE-A Classification: demo-illustrative

Stamp: Research Draft — Not for Construction · DEMO

PINNED DIGESTS (SHA-256)

fixture: b7a5c46d9e58a3dd5758556734e0899eba970296514be7e4ffbe7d079f82e2de

rulebook: fd6c579f1d7fb32b244e192d3021f1eb608804f93a898418b6037c69a36e203f

geometry: 1182bdec34e9d8592db3089f4ad9862c4a9d60ba8d716e52e4c081bc3ae0973

DECLARED FIXTURE INPUTS

Site width (E-W): 225 m [fact.site-width] <- site.widthM

Site depth (N-S): 180 m [fact.site-depth] <- site.depthM

Declared north bearing: 0 deg [fact.north-bearing] <- site.northBearingDeg

External access road width (south): 24 m [fact.access-road-width] <- site.accessRoad.widthM

Requested dwelling units (client intent): 500 DU [fact.requested-du] <- site.requestedDwellingUnits

note: Intent only — never a result.

PERMITTED VALUES (DEMO RULE ENTRIES)

Max ground coverage (of gross site area): 40 percent [fact.rule.site-coverage-max] cites DEMO-A-SITE-COVERAGE-MAX

Max density: 40 du-per-acre [fact.rule.density-max] cites DEMO-A-DENSITY-MAX

Max height: 12 metre [fact.rule.height-max] cites DEMO-A-HEIGHT-MAX

Max storeys: 3 storeys [fact.rule.storeys-max] cites DEMO-A-STOREYS-MAX

Periphery setback: 9 metre [fact.rule.setback-periphery] cites DEMO-A-SETBACK-PERIPHERY

Front (access-road) setback: 12 metre [fact.rule.setback-front] cites DEMO-A-SETBACK-FRONT

Primary internal road width: 12 metre [fact.rule.road-width-primary] cites DEMO-A-ROAD-WIDTH-PRIMARY

Secondary internal road width: 9 metre [fact.rule.road-width-secondary] cites DEMO-A-ROAD-WIDTH-SECONDARY

Min open space (of gross site area): 15 percent [fact.rule.open-space-min] cites DEMO-A-OPEN-SPACE-MIN

Parking norm: 2 ecs-per-du [fact.rule.parking-ecs-per-du] cites DEMO-A-PARKING-ECS-PER-DU

Min amenity share (of gross site area): 5 percent [fact.rule.amenity-share-min] cites DEMO-A-AMENITY-SHARE-MIN

Min townhouse plot frontage: 6 metre [fact.rule.unit-plot-frontage-min] cites DEMO-A-UNIT-PLOT-FRONTAGE-MIN

Min townhouse plot depth: 15 metre [fact.rule.unit-plot-depth-min] cites DEMO-A-UNIT-PLOT-DEPTH-MIN

Max unbroken row length: 72 metre [fact.rule.row-length-max] cites DEMO-A-ROW-LENGTH-MAX

DERIVED ENVELOPE (EVERY NUMBER CITES ITS FEEDERS)

Gross site area: 40500 sqm [fact.site-area]

from fixture: site.widthM, site.depthM

cites: (none — fixture-only derivation)

note: width x depth

Gross site area: 10.008 acre [fact.site-area-acres]

from fixture: site.widthM, site.depthM

cites: (none — fixture-only derivation)

note: site area / 4046.8564224 sqm-per-acre (physical constant)

Buildable envelope width: 207 m [fact.envelope-width]

from fixture: site.widthM

cites: DEMO-A-SETBACK-PERIPHERY

Buildable envelope depth: 159 m [fact.envelope-depth]

from fixture: site.depthM

cites: DEMO-A-SETBACK-FRONT, DEMO-A-SETBACK-PERIPHERY

Ground-coverage cap: 16200 sqm [fact.coverage-cap-area]

from fixture: site.widthM, site.depthM

cites: DEMO-A-SITE-COVERAGE-MAX

note: coverage % x gross site area

Density ceiling: 400 DU [fact.density-ceiling]

from fixture: site.widthM, site.depthM

cites: DEMO-A-DENSITY-MAX

note: floor(site acres x density). A ceiling only — NOT a geometric capacity.

Townhouses placed in this reference layout: 140 DU [fact.placed-du]

from fixture: site.widthM, site.depthM

cites: DEMO-A-SETBACK-PERIPHERY, DEMO-A-SETBACK-FRONT, DEMO-A-ROAD-WIDTH-PRIMARY,
DEMO-A-ROAD-WIDTH-SECONDARY, DEMO-A-OPEN-SPACE-MIN, DEMO-A-AMENITY-SHARE-MIN,
DEMO-A-UNIT-PLOT-FRONTAGE-MIN, DEMO-A-UNIT-PLOT-DEPTH-MIN, DEMO-A-ROW-LENGTH-MAX,

DEMO-A-SITE-COVERAGE-MAX, DEMO-A-DENSITY-MAX

note: Counted from canonical geometry. Not a claimed maximum: no completeness search over the layout space was performed.

Shortfall against request: 360 DU [fact.shortfall-du]

from fixture: site.requestedDwellingUnits, site.widthM, site.depthM

cites: DEMO-A-SETBACK-PERIPHERY, DEMO-A-SETBACK-FRONT, DEMO-A-ROAD-WIDTH-PRIMARY, DEMO-A-ROAD-WIDTH-SECONDARY, DEMO-A-OPEN-SPACE-MIN, DEMO-A-AMENITY-SHARE-MIN, DEMO-A-UNIT-PLOT-FRONTAGE-MIN, DEMO-A-UNIT-PLOT-DEPTH-MIN, DEMO-A-ROW-LENGTH-MAX, DEMO-A-SITE-COVERAGE-MAX, DEMO-A-DENSITY-MAX

note: requested - placed

Built footprint (plots + club): 12780 sqm [fact.built-footprint]

from fixture: site.widthM, site.depthM

cites: DEMO-A-UNIT-PLOT-FRONTAGE-MIN, DEMO-A-UNIT-PLOT-DEPTH-MIN, DEMO-A-AMENITY-SHARE-MIN, DEMO-A-SETBACK-PERIPHERY, DEMO-A-SETBACK-FRONT, DEMO-A-ROAD-WIDTH-PRIMARY, DEMO-A-ROAD-WIDTH-SECONDARY, DEMO-A-OPEN-SPACE-MIN, DEMO-A-AMENITY-SHARE-MIN, DEMO-A-UNIT-PLOT-FRONTAGE-MIN, DEMO-A-UNIT-PLOT-DEPTH-MIN, DEMO-A-ROW-LENGTH-MAX, DEMO-A-SITE-COVERAGE-MAX, DEMO-A-DENSITY-MAX

note: Conservative demo convention: each townhouse plot counted fully as footprint.

Green/open space required: 6075 sqm [fact.green-required]

from fixture: site.widthM, site.depthM

cites: DEMO-A-OPEN-SPACE-MIN

Green/open space provided (measured): 6084 sqm [fact.green-provided]

from fixture: site.widthM, site.depthM

cites: DEMO-A-OPEN-SPACE-MIN, DEMO-A-SETBACK-PERIPHERY, DEMO-A-SETBACK-FRONT, DEMO-A-ROAD-WIDTH-PRIMARY

Amenity parcel required: 2025 sqm [fact.amenity-required]

from fixture: site.widthM, site.depthM

cites: DEMO-A-AMENITY-SHARE-MIN

Amenity parcel provided (measured): 2025 sqm [fact.amenity-provided]

from fixture: site.widthM, site.depthM

cites: DEMO-A-AMENITY-SHARE-MIN, DEMO-A-UNIT-PLOT-DEPTH-MIN

Required parking: 280 ECS [fact.parking-required]

from fixture: site.widthM, site.depthM

cites: DEMO-A-PARKING-ECS-PER-DU, DEMO-A-SETBACK-PERIPHERY, DEMO-A-SETBACK-FRONT, DEMO-A-ROAD-WIDTH-PRIMARY, DEMO-A-ROAD-WIDTH-SECONDARY, DEMO-A-OPEN-SPACE-MIN, DEMO-A-AMENITY-SHARE-MIN, DEMO-A-UNIT-PLOT-FRONTAGE-MIN, DEMO-A-UNIT-PLOT-DEPTH-MIN, DEMO-A-ROW-LENGTH-MAX, DEMO-A-SITE-COVERAGE-MAX, DEMO-A-DENSITY-MAX

note: placed DU x parking norm, rounded up. Parking is on-plot (stilt) in v0 and not drawn.

VERDICT — REQUEST vs CEILING vs PLACED

requested [fact.requested-du] ceiling [fact.density-ceiling]

placed [fact.placed-du] shortfall [fact.shortfall-du]

binding: Geometric fill of this reference layout under the cited setback, road, open-space, amenity, plot-dimension and row-length entries. (DEMO-A-SETBACK-PERIPHERY, DEMO-A-SETBACK-FRONT, DEMO-A-ROAD-WIDTH-PRIMARY, DEMO-A-ROAD-WIDTH-SECONDARY, DEMO-A-OPEN-SPACE-MIN, DEMO-A-AMENITY-SHARE-MIN, DEMO-A-UNIT-PLOT-FRONTAGE-MIN, DEMO-A-UNIT-PLOT-DEPTH-MIN, DEMO-A-ROW-LENGTH-MAX)

The client asked for 500 townhouses. Under the DEMO-SLICE-A entries, the density ceiling is 400

DU and 140 townhouses are placed in this reference layout (shortfall 360). The placed count is what this layout achieves; it is not a claimed legal capacity, because no completeness search over the permitted layout space was performed. All values are illustrative demo entries — this is not a statement about any real jurisdiction.

CITATION SNAPSHOT (SELF-CONTAINED — VALID WITHOUT THE RULEBOOK FILES)

DEMO-A-SITE-COVERAGE-MAX slot=site-coverage-max value=40 percent version=DEMO-A-SITE-COVERAGE-MAX#v1
demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: Built footprint may cover at most 40% of gross site area (denominator: gross site area).

DEMO-A-DENSITY-MAX slot=density-max value=40 du-per-acre version=DEMO-A-DENSITY-MAX#v1

demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: At most 40 dwelling units per

acre of gross site area.

DEMO-A-HEIGHT-MAX slot=height-max value=12 metre version=DEMO-A-HEIGHT-MAX#v1

demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: No structure exceeds 12 m above finished ground level.

DEMO-A-STOREYS-MAX slot=storeys-max value=3 storeys version=DEMO-A-STOREYS-MAX#v1

demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: Townhouses of at most three storeys (G+2).

DEMO-A-SETBACK-PERIPHERY slot=setback-periphery value=9 metre version=DEMO-A-SETBACK-PERIPHERY#v1

demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: A 9 m setback from every site boundary that does not face the access road.

DEMO-A-SETBACK-FRONT slot=setback-front value=12 metre version=DEMO-A-SETBACK-FRONT#v1

demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: A 12 m setback from the boundary facing the external access road.

DEMO-A-ROAD-WIDTH-PRIMARY slot=road-width-primary value=12 metre version=DEMO-A-ROAD-WIDTH-PRIMARY#v1

demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: The internal spine road is 12 m wide, right-of-way.

DEMO-A-ROAD-WIDTH-SECONDARY slot=road-width-secondary value=9 metre version=DEMO-A-ROAD-WIDTH-SECONDARY#v1

demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: Internal secondary/row roads are 9 m wide, right-of-way.

DEMO-A-OPEN-SPACE-MIN slot=open-space-min value=15 percent version=DEMO-A-OPEN-SPACE-MIN#v1

demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: At least 15% of gross site area is green/open space (denominator: gross site area).

DEMO-A-PARKING-ECS-PER-DU slot=parking-ecs-per-du value=2 ecs-per-du version=DEMO-A-PARKING-ECS-PER-DU#v1

demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: Two equivalent car spaces per dwelling unit.

DEMO-A-AMENITY-SHARE-MIN slot=amenity-share-min value=5 percent version=DEMO-A-AMENITY-SHARE-MIN#v1

demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: At least 5% of gross site area is the community amenity parcel, club + pool (denominator: gross site area).

DEMO-A-UNIT-PLOT-FRONTAGE-MIN slot=unit-plot-frontage-min value=6 metre version=DEMO-A-UNIT-PLOT-FRONTAGE-MIN#v1

demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: Each townhouse plot has at least 6 m of road frontage.

DEMO-A-UNIT-PLOT-DEPTH-MIN slot=unit-plot-depth-min value=15 metre version=DEMO-A-UNIT-PLOT-DEPTH-MIN#v1

demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: Each townhouse plot is at least 15 m deep.

DEMO-A-ROW-LENGTH-MAX slot=row-length-max value=72 metre version=DEMO-A-ROW-LENGTH-MAX#v1

demo-illustrative / unverified authority: DEMO AUTHORITY (illustrative)

source: demo — illustrative (no real-world document). basis: One unbroken townhouse row is at most 72 m long.

NOTES

DEMO: every value is illustrative and unverified; no real authority, document, or jurisdiction is represented.

Percent rules use gross site area as denominator, stated per entry; provided areas are measured from canonical geometry.

The requested unit count is client intent and is never copied into any result field.

Height and storey caps are cited envelope limits; v0 draws no elevations, and no drawing label exceeds them.